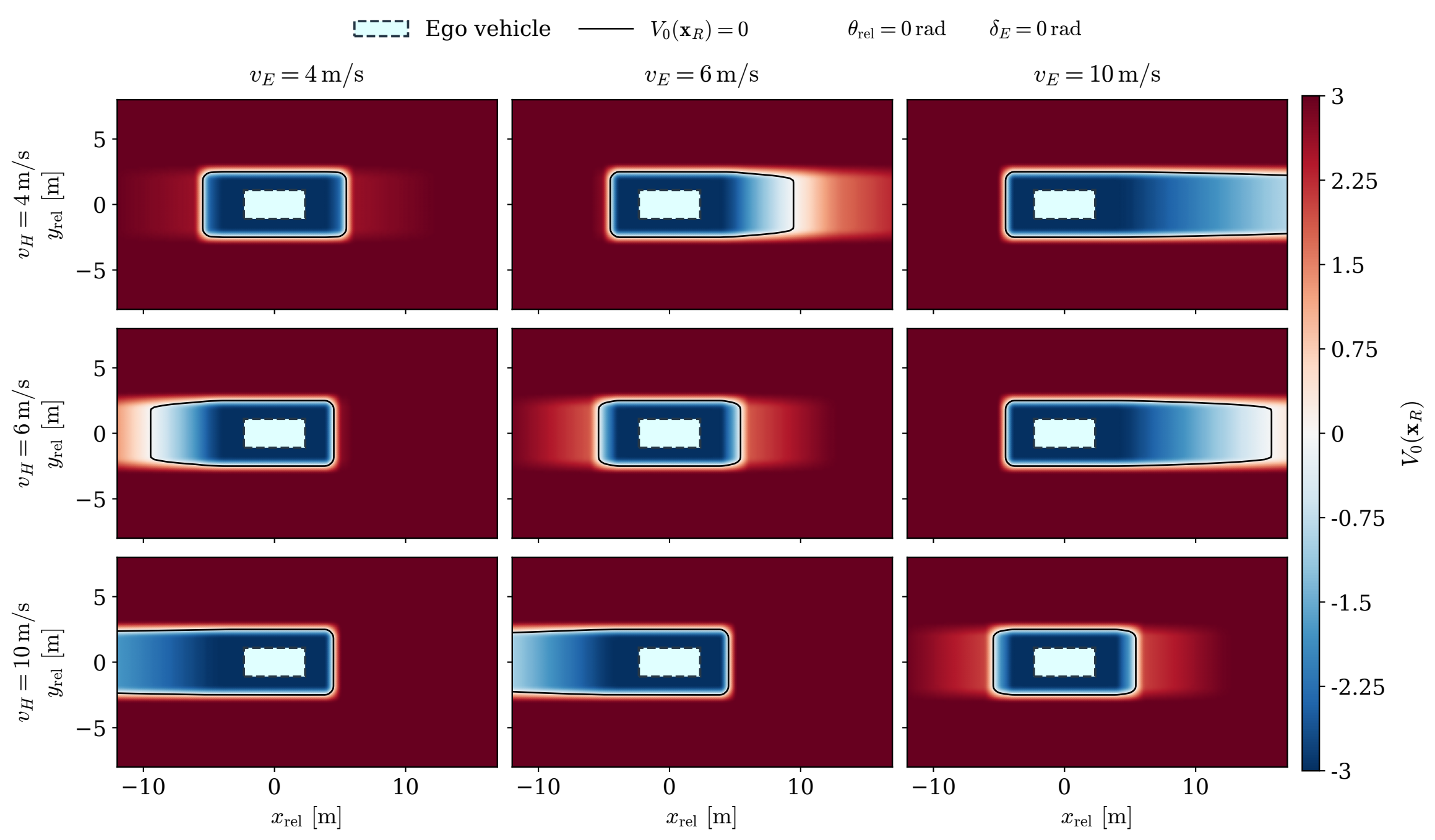

Ego vehicle
$V_0(\mathbf{x}_R) = 0$
$\theta_{\text{rel}} = 0\,\text{rad}$
$\delta_E = 0\,\text{rad}$
$v_E = 4\,\text{m/s}$
$v_E = 6\,\text{m/s}$
$v_E = 10\,\text{m/s}$
$v_H = 4\,\text{m/s}$
$v_H = 6\,\text{m/s}$
$v_H = 10\,\text{m/s}$
y_{rel} [m]
x_{rel} [m]
5
0
−5
−10
0
10
3
2.25
1.5
0.75
0
-0.75
-1.5
-2.25
-3
$V_0(\mathbf{x}_R)$